Holes and Vertical Asymptotes

Factoring and cancelling, telling a hole from a vertical asymptote, and finding the height of a hole

Directions

- Factor the numerator and the denominator completely before deciding anything about the graph.
- A factor that cancels leaves a **hole**: one missing point. A factor still in the denominator after cancelling gives a **vertical asymptote**.
- The height of a hole comes from the *cancelled* form, never from substituting into the original quotient.
- Four problems show work that a student actually handed in. Find the correct answer first, then say what went wrong.

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1. Write $\dfrac{x^2-9}{x^2-x-6}$ in fully simplified form.

Answer: ____________________

2. Let $f(x) = \dfrac{x^2-9}{x^2-x-6}$.

(a) List the values of x that make the *original* denominator zero: _________

(b) One of those values is a hole and the other is a vertical asymptote. Say which is which, and explain what in the factored form tells you.

3. The graph of $f(x) = \dfrac{x^2 - 9}{x^2 - x - 6}$ has exactly one hole. Give the coordinates of that hole as an ordered pair.

Answer: ___________________

4. Let $g(x) = \dfrac{2x^2 - 8}{x^2 + x - 6}$.

(a) Write $g(x)$ in fully simplified form: _________

(b) List every value of x that must be excluded from the domain of g: _________

5. A student was asked for the height of the hole in the graph of $h(x) = \dfrac{x^2 - 4}{x - 2}$ and wrote:

"At $x = 2$ the top is $2^2 - 4 = 0$, so the hole is at height 0."

(a) Find the correct height of the hole: _________

(b) Name the error in the student's reasoning.

6. A student was asked for the vertical asymptotes of $k(x) = \dfrac{x^2 - x - 6}{x^2 - 4}$ and wrote:

"The bottom is zero at $x = 2$ and $x = -2$, so both are vertical asymptotes."

(a) Write $k(x)$ in fully simplified form: __________

(b) Give the equation of the only vertical asymptote: __________

(c) Name the error in the student's reasoning.

7. Let $m(x) = \dfrac{x^3 - x}{x^2 - 1}$.

(a) Write $m(x)$ in fully simplified form: __________

(b) List the x-values at which the graph has a hole: __________

(c) How many vertical asymptotes does the graph have? Explain how the cancelling supports your count.

8. Let $n(x) = \dfrac{x^2 + 2x - 15}{x^2 - 2x - 3}$.

(a) Write $n(x)$ in fully simplified form: _________

(b) Find the height of the hole in the graph of n: _________

(c) Give the equation of the vertical asymptote: _________